



Tushnik Dutta

M.Tech Computer Science
Indian Statistical Institute

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Education

Course	College/University	Year	CGPA/%
M.Tech Computer Science	Indian Statistical Institute	2026–28	–
B.Tech Computer Science Engineering	Government College of Engineering and Leather Technology	2022–26	8.53
Intermediate (+2)	Hariyana Vidya Mandir	2022	95.2%
High School	Hariyana Vidya Mandir	2020	91.4%

Experience

Research Intern, Jio Institute

July 2024 – Oct 2024

Jio Institution of Education and Research ,Navi Mumbai, India

- Worked on deep learning models (CNNs, RNNs, Transformers, AutoEncoders, Diffusion Models) for Brainwave to Image Reconstruction using EEG (Electroencephalogram).
- Implemented a stable diffusion and BLIP model using contrastive loss techniques (Semi-Supervised Learning) to map brain wave patterns to the images, while mentoring postgraduate students as a teaching assistant.

Web Development Training, Internsellite

August 2023 – October 2023

Internsellite ,Hamirpur, Himachal Pradesh, India

- Training on Web Development and Basics.
- Backend Node.js Development.
- Worked on a dummy Bookstore web application.

Projects

LLM Calculator

[Project Link](#)

- A 11 token based calculator identifies only addition, subtraction and multiplication.
- Trained upon dataset containing operands in range of 2 digits and result in atmost 4 digits.
- Limitations: A small model, though follows LLM training procedure, cannot be called one. Accuracy of 2 percent.

RAG-based Medical Prescription Generator

[Project Link](#)

- Built a **Retrieval-Augmented Generation (RAG)** based system that generates medical prescriptions from patient symptoms and diagnostic inputs.
- Future Development: Addition of vector databases to store and quick retrieve diagnostics, for similar symptoms.

Robotics

- **Linobot:** Line-following robot using Arduino.
- **Robo Car with Object Detection:** Use ultrasonic sensors to do real-time object detection.
- **Robo Hand:** Interactive robotic hand prototype demonstrating precise motion control.

Technologies

Programming Languages : C/C++ , Python, R, Javascript

Technologies: git, SQL, LATEX, pytorch, tensorflow, keras

Expertise Areas: Machine Learning, Deep Learning, Image Processing, Computer Vision

Languages: English, Bengali, Hindi

Achievements and Certifications

- **Robotics Student Lead:** Google Developer Club Robotics Leader for year 2025-26.
- **Government Body Certification:** Programming with Python at Government College of Engineering and Ceramic Technology(GCECT)
- **Government Body Certification:** Python for AI, Data Science, Big Data and the Cloud at GCECT
- Secured a **994 All India Rank** in Graduate Aptitude Test in Engineering (**GATE**) in Computer Science in the year 2026.
- Secured a **1718 All India Rank** in Graduate Aptitude Test in Engineering (**GATE**) in Data Science and Artificial Intelligence in the year 2026.

Relevant Courses

- Data Structures and Algorithms
- Database Management Systems
- Operating Systems
- Artificial Intelligence and Machine Learning
- Deep Learning
- Image Processing and Computer Vision
- Natural Language Processing